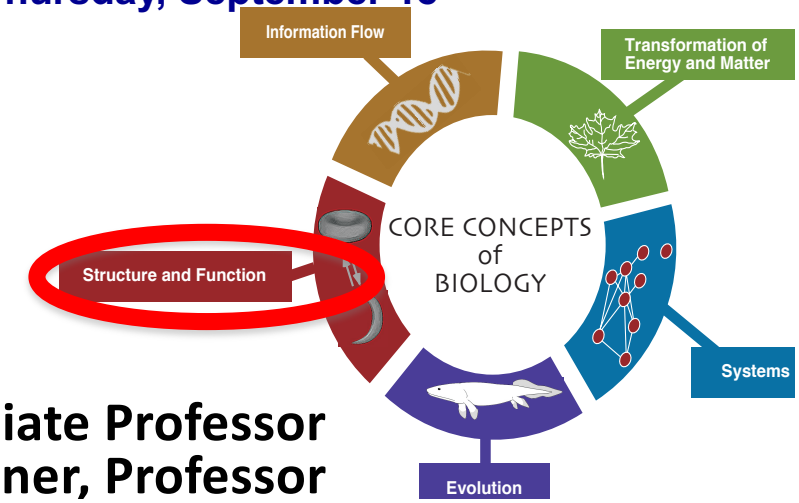
Biol 100 Class #8, Monday, September 15, 2025

Agenda:

How can different kinds of mutations affect the protein produced?
How can a small change in the STRUCTURE of Genes/DNA/Proteins change the FUNCTION and have big impact on human biology?

- **Welcome, Agenda, Announcements, & Checking on U-ABC-IT**
- **Video Animation of Transcription and Translation**
- **Analyzing Gene Puzzle Results:**
  - **Comparing Differences in Genes and Proteins from Activity Becoming RNA Polymerase and Being the Ribosome**
- **Mini-Lecture: Naming Some Types of Gene/DNA Mutations**
- **Clicker Questions: Revisiting Connections between Mutations & Cancer**
- **Last Homework Assignment before Exam #1-**
  - **Reflective Journal #7, due by midnight this Thursday, September 18**





Sally Pasion, Associate Professor
Kimberly Tanner, Professor



Important Announcements!

About Exam #1…

- It is in person during class on
  Next MONDAY, 9/22/25! One week!
  BE HERE EARLY!
- It will be a mix of multiple choice, essay, & extra credit questions.
- It will address things we've learned from Class #1 through #10.

Go Team Bio 100!

Preparing for Exam #1…

- Please DO study some every day.
- Please DO study with colleagues, if possible.
  – Look around you now! Look for Bag Tags!
- Please DO use the Biology 100 Exam #1 Study Guide on Canvas!
- Please DO learn for the long term and NOT the short term.
- Please do NOT wait to cram two days before; it doesn't work…



Name Tent Reflection…

Please THINK and WRITE…

What are the 6 habits of mind we focus on practicing in this Human Biology course?

1._____

2._____

3._____

4._____

5._____

6._____

Please leave all name tents in the bin at your table after EVERY class session!



Self-Assessment…

Clicker Question

The product of the process of _____ is a new _____ molecule.

A) transcription…DNA

B) transcription…protein

C) translation…RNA

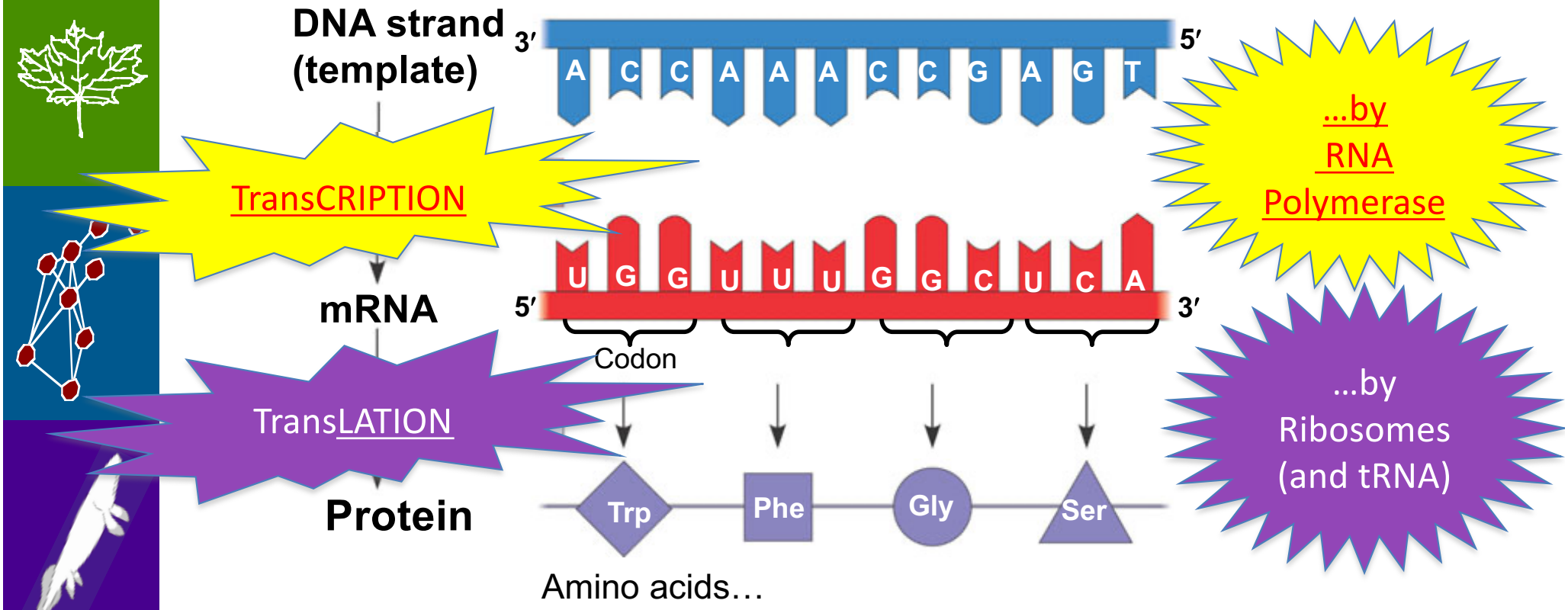
D) translation…protein

E) DNA replication…RNA

Activity: Becoming RNA Polymerase and Being a Ribosome

You did transcription and translation on paper for multiple versions (ALLELES) of a Mystery Gene! Like a series of puzzles….





DNA (genes) → RNA → Proteins → Traits



3-minute video may help you build a model for TRANSCRIPTION and TRANSLATION in your mind!

Getting from DNA/Genes to Proteins/Traits –
Transcription by RNA Polymerase
And
Translation by Ribosomes
https://www.youtube.com/watch?v=nHM4UUVHPQM



Self-Assessment…

Clicker Question

The product of the process of _____ is a new _____ molecule.

A) transcription…DNA

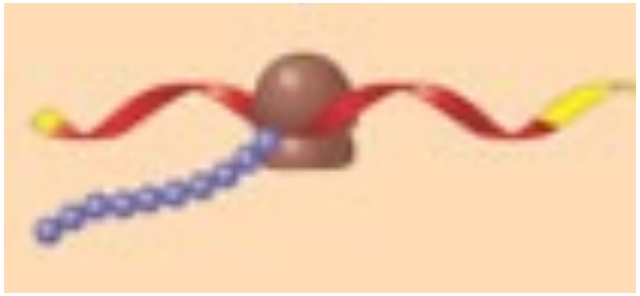
B) transcription…protein

C) translation…RNA

D) translation…protein

E) DNA replication…RNA

Activity: Becoming RNA Polymerase and Being the Ribosome…



Clicker Question

Which of the different alleles (versions) of the Mystery Gene produced DIFFERENT PROTEINS compared to the Normal Mystery Gene?

A) All of them

B) None of them

C) Only Allele B

D) Allele B and Allele C

E) Allele B, Allele D, and Allele E



What is different about Allele B?!?

Template DNA Sequences

Allele A 3'TAC CAA GTA GAC TGA GGA CTT CTT…5'
Allele B 3'TAC CAA GTA GAC TGA GGA CTT CAT…5'

Allele A = "Normal" Allele

Transcribed RNA Sequences

Allele A 5'AUG GUU CAU CUG ACU CCU GAA GAA…3'
Allele B 5'AUG GUU CAU CUG ACU CCU GAA GUA…3'

Translated Protein

Allele A MET-VAL-HIS-LEU-THR-PRO-GLU-GLU…
Allele B MET-VAL-HIS-LEU-THR-PRO-GLU-VAL…

Tip for looking at all these letters!

Ask yourself…

- *Is the protein different? How?*
- *Is the RNA different? How?*
- *Is the DNA different? How?*



What is different about Allele B?!?

Template DNA Sequences

Allele A 3'TAC CAA GTA GAC TGA GGA CTT CTT…5'
Allele B 3'TAC CAA GTA GAC TGA GGA CTT C**A**T…5'

Allele A = "Normal" Allele

Transcribed RNA Sequences

Allele A 5'AUG GUU CAU CUG ACU CCU GAA GAA…3'
Allele B 5'AUG GUU CAU CUG ACU CCU GAA G**U**A…3'

Translated Protein

Allele A MET-VAL-HIS-LEU-THR-PRO-GLU-GLU…
Allele B MET-VAL-HIS-LEU-THR-PRO-GLU-**VAL**…

Tip for looking at all these letters!

Ask yourself…

- *Is the protein different? How?*
- *Is the RNA different? How?*
- *Is the DNA different? How?*

SUBSTITUTION Mutation
MISSENSE Mutation
(different protein made >>> Sickle Cell Mutation)

Mystery Gene=Hemoglobin Gene!

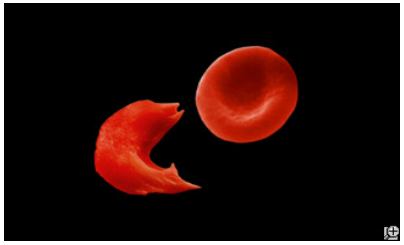
Sickle Cell Disease/Anemia

Allele B Substitution Mutation, Missense



Tionne Watkins aka T-Boz from music group TLC is one of many ambassadors for promoting understanding of Sickle Cell

- Mutation in hemoglobin gene that codes for hemoglobin protein, which carries oxygen around the body
- Gene is transcribed (copied) and protein is translated (built) in red blood cells before they mature
- Causes “sickling” change of shape of red blood cells
- Variety of physiological effects



- Happens when you inherit mutation from both parents
- More common in African American and Mediterranean populations

Another resource…
Sickle Cell 101Non-Profit
https://www.sc101.org



What is different about Allele C?!?

Template DNA Sequences

Allele A 3'TAC CAA GTA GAC TGA GGA CTT CTT…5'
Allele C 3'TAC CAA GTA GAC TGG GGA CTT CTT…5'

Allele A = "Normal" Allele

Transcribed RNA Sequences

Allele A 5'AUG GUU CAU CUG ACU CCU GAA GAA…3'
Allele C 5'AUG GUU CAU CUG ACC CCU GAA GAA…3'

Translated Protein

Allele A MET-VAL-HIS-LEU-THR-PRO-GLU-GLU…
Allele C MET-VAL-HIS-LEU-THR-PRO-GLU-GLU…

Remember!
Codon= triplet nucleotides in RNA that code for a single amino acid

Tip for looking at all these letters!

Ask yourself…

- *Is the protein different? How?*
- *Is the RNA different? How?*
- *Is the DNA different? How?*



What is different about Allele C?!?

Template DNA Sequences

Allele A 3'TAC CAA GTA GAC TGA GGA CTT CTT…5'
Allele C 3'TAC CAA GTA GAC TGG GGA CTT CTT…5'

Allele A = "Normal" Allele

Transcribed RNA Sequences

Allele A 5'AUG GUU CAU CUG ACU CCU GAA GAA…3'
Allele C 5'AUG GUU CAU CUG ACC CUU GAA GAA…3'

Translated Protein

Allele A MET-VAL-HIS-LEU-THR-PRO-GLU-GLU…
Allele C MET-VAL-HIS-LEU-THR-PRO-GLU-GLU…

Tip for looking at all these letters!

Ask yourself…

- *Is the protein different? How?*
- *Is the RNA different? How?*
- *Is the DNA different? How?*

Substitution Mutation
SENSE Mutation
SILENT Mutation
(same protein made!)



What is different about Alleles D & E?!?

Allele A = "Normal" Allele

Template DNA Sequences

Allele A 3'TAC CAA GTA GAC TGA GGA CTT CTT…5'
Allele D 3'TAC CAA AGT AGA CTG AGG ACT TCT T…5'
Allele E 3'TAC CAA GTA GAT GAG GAC TTC TT…5'

Transcribed RNA Sequences

Allele A 5'AUG GUU CAU CUG ACU CCU GAA GAA…3'
Allele D 5'AUG GUU UCA UCU GAC UCC UGA AGA A…3'
Allele E 5'AUG GUU CAU CUA CUC CUG AAG AA…3'

Translated Protein

Allele A MET-VAL-HIS-LEU-THR-PRO-GLU-GLU…
Allele D MET-VAL-SER-SER-ASP-SER-STOP!
Allele E MET-VAL-HIS-LEU-LEU-LEU-LYS…

Tip for looking at all these letters!

Ask yourself…

- *Is the protein different? How?*
- *Is the RNA different? How?*
- *Is the DNA different? How?*



What is different about Alleles D & E?!?

Allele A = "Normal" Allele

Template DNA Sequences

Allele A 3'TAC CAA GTA GAC TGA GGA CTT CTT…5'
Allele D 3'TAC CAA AGT AGA CTG AGG ACT TCT T…5'
Allele E 3'TAC CAA GTA GA_T GAG GAC TTC TT…5'

Transcribed RNA Sequences

Allele A 5'AUG GUU CAU CUG ACU CCU GAA GAA…3'
Allele D 5'AUG GUU UCA UCU GAC UCC UGA AGA A…3'
Allele E 5'AUG GUU CAU CU_ A CUC CUG AAG AA…3'

Translated Protein

Allele A MET-VAL-HIS-LEU-THR-PRO-GLU-GLU…
Allele D MET-VAL-**SER-SER-ASP-SER-STOP!**
Allele E MET-VAL-HIS-LEU-**LEU-LEU-LYS…**

Tip for looking at all these letters!

Ask yourself…

- *Is the protein different? How?*
- *Is the RNA different? How?*
- *Is the DNA different? How?*

Insertion (Allele D) or Deletion (Allele E) Mutation

Frameshift Mutations!

ALWAYS causes different protein to be made (if at all)!



Naming Some Types of Mutations

Allele B Substitution Mutation, Missense (different protein; Sickle Cell Mutation!)

Allele C Substitution Mutation, Silent, Sense (same protein made)

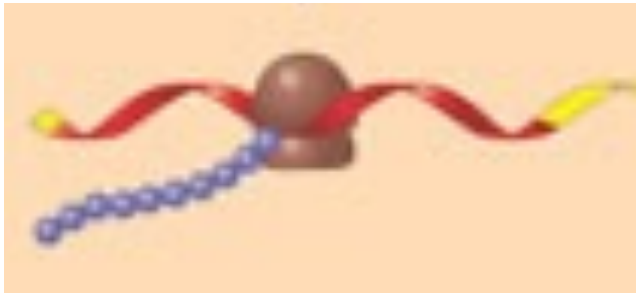
Allele D Insertion Mutation, Frameshift! different protein made

Allele E Deletion Mutation, Frameshift! different protein made

MUTATIONS= changes in the nucleotide sequence of DNA

Allele A = "Normal" Allele

Activity: Becoming RNA Polymerase and Being the Ribosome…



Clicker Question

Which of the different alleles (versions) of the Mystery Gene produced DIFFERENT PROTEINS compared to the Normal Mystery Gene?

A) All of them

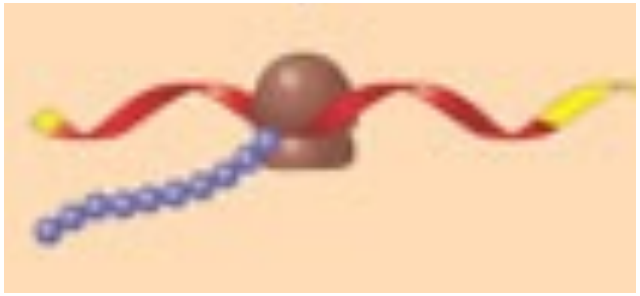
B) None of them

C) Only Allele B

D) Allele B and Allele C

E) Allele B, Allele D, and Allele E

Activity: Becoming RNA Polymerase and Being the Ribosome…



Clicker Question

Which of the different alleles (versions) of the Mystery Gene might you choose for your future child, if they did not inherit the "normal" allele?

A) Allele B

B) Allele C

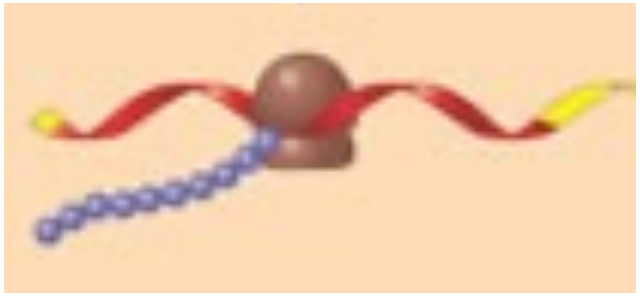
C) Allele D

D) Allele E

E) Any of these Alleles would be fine…



Activity: Becoming RNA Polymerase and Being the Ribosome…



Clicker Question

Which of the following kinds of **single nucleotide mutations** in a gene ***will always produce a different protein*** compared to the protein produced by the normal version of the gene?

A. only a **substitution** mutation
B. only a **deletion** mutation
C. only an **insertion** mutation
D. either a **deletion** or an **insertion** mutation
E. either a **substitution** or an **insertion** mutation



Summary of Types of Mutations

Allele B — **Substitution Mutation, Missense** (different protein; Sickle Cell Mutation!)

Allele C — **Substitution Mutation, Silent, Sense** (same protein made)

Allele D — **Insertion Mutation, Frameshift!** different protein made

Allele E — **Deletion Mutation, Frameshift!** different protein made

Insertion and deletion mutations (if only one nucleotide) always result in a frameshift and therefore a different protein product!

Not all mutations involve just one nucleotide



Activity: Becoming RNA Polymerase and Being a Ribosome

See Canvas for file with the RNA and Protein solutions for these Gene Puzzles!



BIG IDEA–
Changing the STRUCTURE of Genes/DNA/Proteins Changes their FUNCTION and resulting Traits

Sickle Cell Anemia

Biotechnology

CANCER

So much more…



Assignments

Due Thursday, September 18 by Midnight (11:59pm)

- **Complete Reflective Journal #7 –**
  Practice Exam #1 Questions – Applying What You've Learned
  See specific instructions on Canvas.

Still open from Sunday, September 14 to Saturday, September 20 at Midnight (11:59pm)

- **Self-Assessment Quiz #3, see Canvas, DUE DATE EXTENDED a WEEK!**
- **Also, please use the PRACTICE versions of Self-Assessment Quizzes #1-2 posted on Canvas to study.**

Please submit your work on time, as late work cannot be graded after the 48-hour grade period. See syllabus.

OPTIONAL Videos: These very short overviews may be helpful:
http://learn.genetics.utah.edu/content/basics/

What are traits?
What are genes?
What are proteins?
Transcribe and translate a gene!
What is a mutation?

Please submit your work on time, as late work cannot be graded after the 48-hour grade period.



Go Team Bio 100!